

WHAT'S NEW IN INTENSIVE CARE



Bedside-available strategies to minimise P-SILI and VILI during ARDS

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Ventilator-induced (VILI) and patient-self-induced lung injury (P-SILI) contribute to acute respiratory distress syndrome (ARDS) progression in a relevant proportion of patients. VILI is a well-established concept encompassing all deleterious effects of mechanical ventilation (MV) [1]. It is a consequence of the excessive stress (pressure applied to the lungs) and strain (deformation induced by inflated volume) imposed by MV to the aerated lung. VILI jeopardises patient's outcome through lung biotrauma, which induces systemic inflammation and may contribute to multiorgan failure, a frequent cause of death in ARDS patients. P-SILI is a more recent concept and includes all the conditions in which spontaneous breathing in ARDS patients exhibiting high inspiratory effort causes the inflation of large tidal volumes (V_t) yielding large transpulmonary pressure (P_L) swings, increases in transvascular pressure favouring hydrostatic pulmonary oedema and regional overinflation through an intra-tidal redistribution of V_t from non-dependent to dependent lung regions (i.e. pendelluft) [2].

We summarise the strategies that can potentially prevent the physiological mechanisms of both P-SILI and VILI at the bedside (Fig. 1). These are based on patient monitoring to provide personalised treatments tailored to physiological phenotypes. Most of these can be applied without additional equipment and may be feasible in resource-limited settings.

Safer use of noninvasive respiratory support

Different noninvasive respiratory support techniques have distinct physiological effects that may influence

outcomes [3]. High-flow nasal oxygen (HFNO) reduces the risk of intubation in acute hypoxemic respiratory failure (AHRF) [3]. CPAP and noninvasive ventilation, despite being applied worldwide, are not formally recommended but can be considered. Prolonged sessions of awake prone position may provide further advantages on lung protection in certain clinical circumstances.

Independently from the chosen technique, delayed intubation has been associated with worse outcomes [4], possibly because of the occurrence of P-SILI during treatment. The time of exposure to hypoxemia and elevated respiratory rates (RR) during noninvasive support is proportional to respiratory system compliance (Crs) deterioration, which makes it difficult to maintain gas exchange, keeping driving pressure (ΔP) within safe limits after intubation [5]. Accordingly, patients' monitoring for early detection of noninvasive support failure is essential during any treatment. Worsening Respiratory rate-Oxygenation (ROX) index, defined as the ratio between SpO_2/FiO_2 and respiratory rate (RR), and heart rate, acidosis, consciousness, oxygenation, and respiratory rate (HACOR) score, consisting of 5 different variables (heart rate, acidosis [pH], consciousness [GCS], oxygenation, and RR) are reliable tools to identify HFNO and continuous positive airway pressure (CPAP) (ROX) and noninvasive ventilation (NIV) (HACOR) failure [4]. Another accurate predictor of NIV failure is the lack of inspiratory effort reduction, which, however, requires oesophageal pressure monitoring [6]. This might not be feasible in all patients with AHRF, and alternative techniques to noninvasively estimate inspiratory effort are being developed. Finally, clinical signs such as worsening PaO_2/FiO_2 ratio, respiratory rate, and dyspnoea are indicative of the patient's deterioration and may prompt the decision to intubate.

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Safer use of controlled mechanical ventilation

Tidal volume and driving pressure

Strain and stress are the major determinants of VILI.

ΔP refers to the stress applied to the lung during tidal ventilation. It is the ratio between V_t and C_{rs} , which is proportional to the size of the aerated lung. Patients with high respiratory system elastance (E_{rs}) (>3 cmH₂O/(mL/kg)) are most likely to benefit from lower V_t [9]. In contrast, allowing $V_t > 6$ mL/kg of predicted body weight (PBW) in patients with low elastance allows less sedation and facilitates the transition to assisted ventilation. Therefore, a pragmatic approach could be starting with 6 mL/kg PBW, and then titrating V_t to achieve a $\Delta P < 15$ cmH₂O. Significantly, PEEP below the airway opening pressure (AOP) can substantially overestimate ΔP . Ongoing trials aiming to test whether titrating V_t limiting ΔP are underway (NCT05440851). Strain represents the deformation of lung units during tidal ventilation, and it is defined as the ratio between V_t and the end-expiratory lung volume (the lung size at PEEP). Traditionally, V_t could be delivered using volume or pressure-controlled ventilation, as no difference in outcomes has been observed [7]. Moreover, it has been suggested that airway pressure release ventilation using time-controlled adaptive ventilation may be even more protective in terms of VILI [8].

Finally, the extent of V_t reduction to minimise ΔP should, however, be balanced to the increase in RR needed to ensure adequate alveolar ventilation, since high RR is itself injurious. Clinically, the combination of V_t and RR that minimises the sum of $[(4 \times \Delta P) + RR]$ best limits the risk of death [10]. Mechanical power was also described aiming to include the effect of RR and PEEP on the effect of airway pressure and V_t in VILI development [11]. Despite being physiologically attractive, the main limitation of its use is that it is not easily calculated at the bedside.

The use of prone position

In intubated patients, sessions of prone position for 12–18 h in moderate-to-severe ARDS patients convincingly reduce the risk of VILI and reverse right ventricle failure and are a simple intervention to improve oxygenation [12]. To prevent VILI, the prone position should be maintained regardless of the improvement in oxygenation achieved.

PEEP and recruitability

High PEEP prevents VILI only if it increases the volume of the aerated lung to an extent capable of sufficiently reducing dynamic strain (strain during tidal ventilation). This varies according to different degrees of recruitability. High PEEP in patients with poor recruitability results in baby lung overinflation and VILI. Recruitability is difficult to assess at the bedside. Computed tomography scan is the gold standard, but may not always be clinically feasible. Different bedside methods to assess recruitability have been described.

First, the recruitment-to-inflation (R/I) ratio can be measured through a single-breath PEEP-reduction manoeuvre with a low PEEP level above the AOP [13]. Importantly, the R/I ratio accurately estimates the extent of PEEP-induced reduction in dynamic strain. Low R/I ratio values (<0.5 – 0.3) accurately identify patients with poor PEEP-induced recruitment and may benefit from lower PEEP. Conversely, high PEEP might be considered in patients exhibiting high R/I values (>0.5 – 0.7).

Second, the hysteresis ratio, i.e. the maximal distance between the inspiratory and expiratory limb weighted by the maximal volume generated during a low-flow pressure–volume [14], and the C_{rs} changes during a decremental PEEP trial have also been shown to predict lung recruitment.

Some imaging techniques could be used. Electrical impedance tomography monitoring during a decremental PEEP trial allows identifying the level of PEEP that

(See figure on next page.)

Fig. 1 Summary of the different strategies that could be applied at the bedside in ARDS patients to minimise the risk of VILI and P-SILI. **a** Relation of V_t with the product of four times driving pressure and respiratory rate at different levels of respiratory system compliance; **b** pressure–volumes curves at different levels of PEEP to estimate the PEEP-induced recruited volume; **c** recruitment-to-inflation manoeuvre based on a single-breath PEEP decrease and estimation of the exhaled tidal volume in patients with different levels of recruitability; **d** estimation of transpulmonary driving using oesophageal pressure measurement; **e** decremental PEEP strategy with electrical impedance tomography to estimate the point with the better balance between collapse and overdistension; **f** expiratory end-occlusion manoeuvre to estimate the occlusion pressure (P_{occ}); end-inspiratory occlusion manoeuvre to estimate the airway driving pressure and the plateau pressure. *HFNO* high-flow nasal oxygen, *CPAP* continuous positive airway pressure, *NIV* noninvasive ventilation, V_t tidal volume, ΔP driving pressure, *RR* respiratory rate, E_{rs} elastance of the respiratory system, *PBW* predicted body weight, P_{plat} plateau pressure, P_L transpulmonary pressure, *HME* heat and moisture exchangers, *PP* prone position, *PEEP* positive end-expiratory pressure, *PV* pressure–volume, *EIT* electrical impedance tomography, *R/I* recruitment to inflation, P_{occ} airway pressure generated by respiratory muscle effort during an expiratory occlusion, ΔP_{Ldyn} dynamic transpulmonary driving pressure, FiO_2 fraction of inspired oxygen

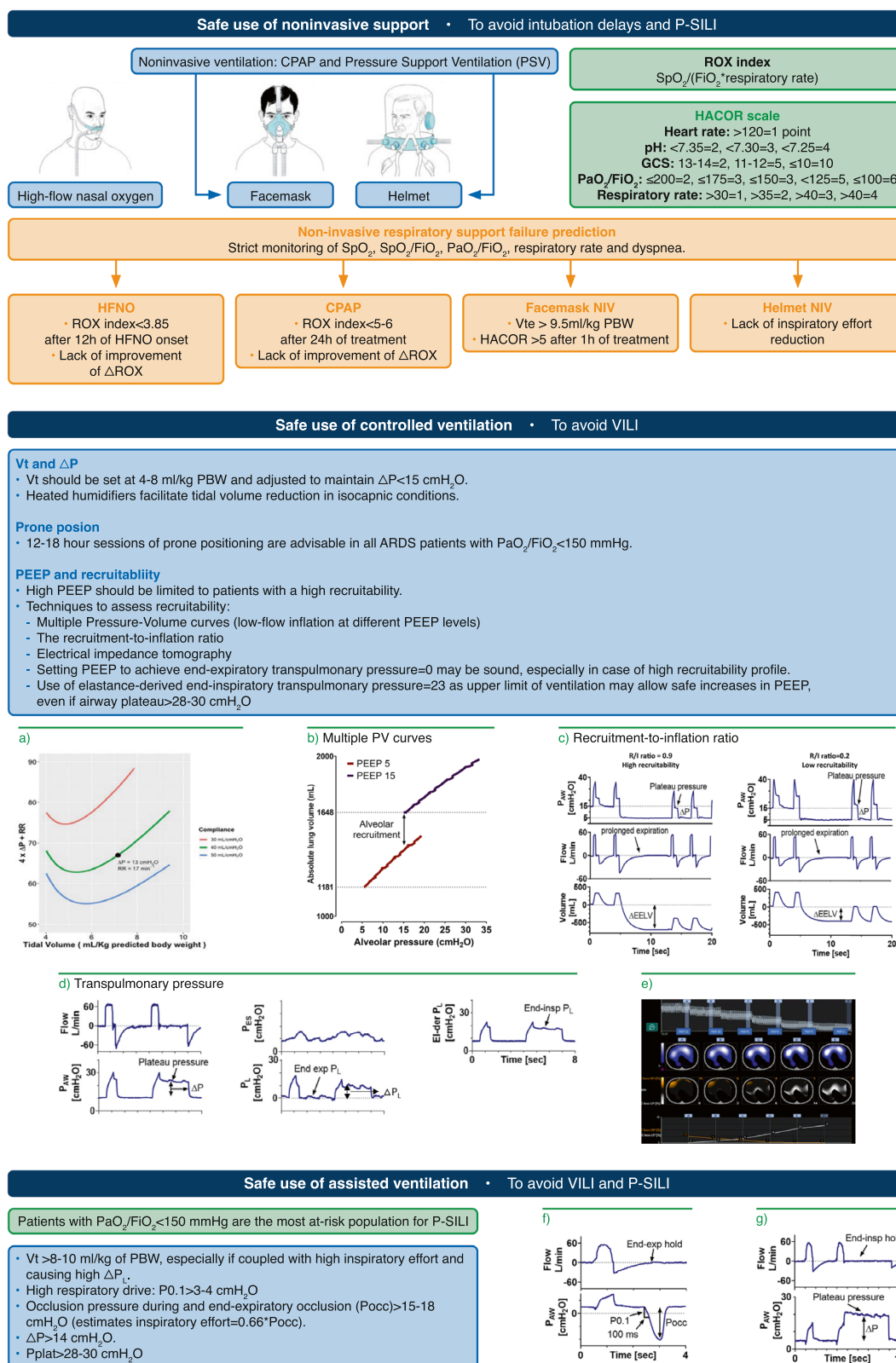


Fig. 1 (See legend on previous page.)

best compromises regional collapse and overdistension. This approach seems effective in setting PEEP based on recruitability and confirmed that PEEP-induced changes in Crs (alone) are poorly informative of recruitment [15]. All these strategies may aid PEEP individualisation at the bedside. Moreover, although the bedside assessment of biological subphenotypes is not feasible, hyperinflammatory ARDS patients may benefit from higher PEEP [16].

Finally, recent ARDS guidelines recommend against the use of prolonged recruitment manoeuvres (RM) [3]. However, the question of whether brief RM in highly recruitable patients or the application of sigh during pressure support ventilation may be associated with some benefit remains unanswered.

Transpulmonary pressure and end-inspiratory lung stress

Transpulmonary pressure (P_L) represents lung stress which, together with strain, is a major determinant of VILI [17]. Absolute values of oesophageal pressure (P_{es}) accurately reflect pleural pressure in the mid-to-dependent zones of the lung. This represents the lung superimposed pressure contributing to lung collapse [18]. End-expiratory P_L (P_{Lee}) is calculated as PEEP—end-expiratory P_{es} during an expiratory pause. It has been shown that setting PEEP to obtain end-expiratory P_L close to 0 to avoid atelectrauma may improve survival [19].

Using P_L , the static end-inspiratory lung stress could be estimated by direct measurements or the elastance-derived method. The direct value of ΔP_L can be obtained using the formula end-inspiratory P_L (P_{Lei}) – P_{Lee} . The elastance-derived method is based on the fact that, because of a ventral-to-dorsal gravitational gradient, P_{es} does not reliably estimate pleural pressure value in non-dependent lung regions most susceptible to overdistension. P_{Lei} can be estimated by plateau pressure (P_{plat}) times the ratio of lung to respiratory system elastance (E_L/E_{RS}) [18]. A safe limit for P_{Lei} may be 20–23 cmH₂O. This may have important clinical implications: first, in patients with high E_L/E_{RS} ratio, P_{plat} of 28–30 cmH₂O is only apparently safe and should be reduced. In patients with low E_L/E_{RS} , it could be safe to increase PEEP (if the patient is recruitable) and tolerate P_{plat} exceeding 30 cmH₂O, with benefits on oxygenation [20].

Safer use of assisted modes and spontaneous breathing

In ARDS, assisted ventilation should be resumed as soon as clinically possible. During assisted ventilation, inspiratory effort should be monitored to minimise the risk of P-SILI. Although there is no conclusive safe threshold, inspiratory effort should not exceed 10–15 cmH₂O. If oesophageal manometry is unavailable, a simple

manoeuvre can help assess inspiratory effort: 75% of the deflection in airway pressure generated by the first effort during an expiratory hold accurately reflects inspiratory effort.

ΔP can also be measured in most patients during assisted ventilation, and safe limits are consistent with what was previously reported. Regarding PEEP, its increase reduces the inspiratory effort when there is an increase in respiratory system compliance [21].

Airway occlusion pressure ($P_{0.1}$) is a measure of respiratory drive, and values >3.5–4 cmH₂O suggest injurious inspiratory effort [22]. High respiratory drive and high inspiratory efforts can lead to asynchronies, such as breath stacking, that may be associated with lung injury [23].

Proper pressure support and sedation selection may help modulate inspiratory effort and respiratory drive. Propofol and opioids mostly reduce the effort and respiratory rate, respectively.

Finally, phrenic nerve stimulation may facilitate the transition to assisted breathing while decreasing the risk of VILI, as it has been associated with positive effects on lung mechanics [24]. Similarly, prone position during spontaneous breathing results in less effort-dependent lung injury [25].

Take-home message

Integrative physiological monitoring of ARDS patients may help bedside personalised treatments to minimise the risk of lung injury progression.

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Declarations

Conflicts of interest

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